

SLANGKOP, South Africa

WMO: 689120

Lat: 34.1483S

Lon: 18.3192E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	10.1	1.8	4.3	17.8	3.7	4.9	14.4	17.0	13.7	15.3	13.6	2.2	140	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	26.0	17.8	24.6	17.6	23.3	17.1	19.6	23.6	18.9	22.6	18.3	21.7	6.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.1	21.5	17.6	12.6	20.7	17.0	12.1	20.1	56.1	23.7	53.7	22.7	51.9	21.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.9	11.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	18.8	18.9	17.8	16.6	15.4	14.4	13.9	13.6	14.3	15.8	16.6	18.2		
	DBStd	2.65	1.92	2.04	1.98	2.29	1.80	1.65	2.06	1.52	1.73	1.98	1.99	1.95		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	917	16	14	34	62	94	118	139	146	124	84	59	27		
	CDD10.0	2257	273	249	242	198	167	133	122	113	128	179	199	253		
	CDD18.3	132	31	31	18	10	3	1	2	1	2	4	8	22		
	CDH23.3	283	65	59	36	34	13	3	6	1	7	9	14	36		
	CDH26.7	27	4	6	4	6	1	0	0	0	2	0	1	4		
(14) Wind	WSAvg	5.3	5.9	5.3	5.0	4.5	4.5	5.6	5.2	5.3	5.3	5.2	5.6	5.7		
(15) Precipitation	PrecAvg	703	22	16	22	54	95	106	117	115	67	36	34	21		
	PrecMax	919	98	36	75	129	175	185	252	206	160	129	100	45		
	PrecMin	465	0	0	0	12	6	29	44	53	22	0	2	0		
	PrecStd	127	22	10	19	28	38	33	50	48	34	25	27	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	27.1	26.8	28.1	25.1	23.0	24.2	22.4	23.9	25.1	25.2	26.1		
		MCWB	19.2	19.5	18.4	16.6	15.5	13.8	13.3	11.8	14.7	16.1	16.3	19.0		
	2%	DB	25.2	25.0	23.9	23.6	21.2	19.6	21.5	18.8	19.3	22.0	22.7	24.1		
		MCWB	19.0	18.7	17.6	16.5	14.9	12.9	12.2	13.1	14.0	15.7	16.2	18.1		
	5%	DB	23.4	23.5	22.3	21.2	19.2	18.0	18.8	17.1	17.7	20.3	20.9	22.5		
		MCWB	18.0	18.3	17.2	16.0	14.6	13.7	13.0	13.1	13.7	15.2	15.6	17.3		
	10%	DB	22.0	21.9	21.0	19.8	17.9	16.8	16.7	16.0	16.5	18.9	19.6	21.2		
		MCWB	17.6	17.6	16.8	15.6	14.6	13.5	13.1	12.9	13.3	14.8	15.3	16.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	20.8	19.7	18.3	17.1	15.9	15.1	15.2	15.7	17.4	18.0	19.8		
		MCDB	25.2	24.8	23.1	24.4	19.5	17.8	17.9	18.0	19.8	21.8	22.5	24.5		
	2%	WB	19.6	19.6	18.6	17.5	16.2	15.1	14.5	14.4	14.7	16.5	16.9	18.9		
		MCDB	23.5	23.3	21.8	21.2	18.5	17.0	17.1	16.7	17.7	20.4	20.5	22.8		
	5%	WB	18.7	18.8	18.0	16.8	15.6	14.6	14.0	13.7	14.2	15.7	16.3	18.1		
		MCDB	22.3	22.1	21.3	19.8	17.9	16.4	16.4	15.8	16.7	18.9	19.6	21.4		
	10%	WB	18.1	18.2	17.4	16.2	15.1	14.1	13.6	13.2	13.7	15.1	15.7	17.5		
		MCDB	21.3	21.3	20.3	18.9	17.1	16.0	15.9	15.2	16.0	17.9	18.8	20.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.1	5.0	4.9	4.4	4.1	4.3	4.0	4.1	4.5	4.5	4.9		
		MCDBR	7.6	7.6	7.6	7.5	7.0	6.5	8.1	6.2	6.5	6.9	6.9	7.0		
	5% WB	MCWBR	3.9	3.8	3.6	3.6	3.5	3.8	3.7	4.0	3.6	3.6	3.5	3.7		
		MCWBR	6.5	6.5	6.0	5.8	5.2	4.5	5.2	4.9	5.0	5.4	5.8	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.348	0.337	0.326	0.325	0.320	0.321	0.335	0.345	0.355	0.342	0.347		
		taud	2.554	2.557	2.582	2.607	2.597	2.584	2.576	2.536	2.470	2.460	2.538	2.539		
	Ebn at Noon		991	968	943	899	843	820	838	873	918	949	988	993		
		Edh at Noon	108	104	94	82	73	70	74	86	104	113	109	110		
(44) All-Sky Solar Radiation	RadAvg		8.11	7.22	5.77	4.22	2.93	2.46	2.74	3.50	4.84	6.37	7.58	8.23		
	RadStd		0.19	0.22	0.23	0.21	0.18	0.14	0.16	0.17	0.19	0.29	0.23	0.19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61	

Nomenclature: See separate page